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五种商品意大利面在碱水煮制下的尺寸变化、时间进程与中式面条适配性：一项探索性厨房比较

Dimensional Changes, Time Course, and Chinese-Noodle Applications of Five Commercial Pasta Types Cooked in Alkaline Water: An Exploratory Kitchen Comparison

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语言 / Language: 中文、English、Bilingual Chinese-English　许可 / License: CC BY 4.0



Figure 1. Finished dish and 2.8 mm cooked-spaghetti measurement.

摘要与关键词 / Abstract and Keywords

摘要： 在 1 L 沸水加入 4 g 小苏打的统一条件下，本研究比较 Angel Hair、Thin Spaghetti、Spaghetti、Linguine 和 Fettuccine。建议煮制时间为 5、9、15、17 和 18 分钟；平均尺寸膨胀倍数为 1.50。

Abstract: Five commercial pasta types were compared under 4 g baking soda in 1 L boiling water. Recommended cooking times were 5, 9, 15, 17, and 18 minutes. The mean dimensional expansion ratio was 1.50.

关键词 / Keywords: 碱水煮制；卡尺测量；尺寸膨胀；意大利面 / alkaline cooking; caliper measurement; dimensional expansion; pasta

1. 背景 / Background

本研究比较五种规格在统一碱水条件下的尺寸变化、感官端点和中式面条应用。 / This study compared size change, sensory endpoint, and possible Chinese-noodle applications across five pasta formats.

2. 材料与方法 / Materials and Methods

所有产品由作者自行购买。每种面条在 1 L 沸水加 4 g 小苏打条件下煮制，达到作者判断的刚好弹牙状态后记录时间，并用数显卡尺测量。圆面记录直径，扁面记录宽度。 / Products were independently purchased. Each was cooked in 1 L boiling water with 4 g baking soda. Time was recorded at the author-defined al dente endpoint. Diameter was measured for round strands and width for flat pasta.



Figure 2. Five commercial pasta products.

3. 结果 / Results

Type	Dry	Cooked	Ratio	Time
Angel Hair	1.2	1.8	1.50x	5
Thin Spag.	1.4	2.1	1.50x	9
Spaghetti	1.8	2.8	1.56x	15
Linguine*	3.1	4.6	1.48x	17
Fettuccine*	4.6	6.8	1.48x	18

* Flat-pasta values are widths; round-pasta values are diameters.

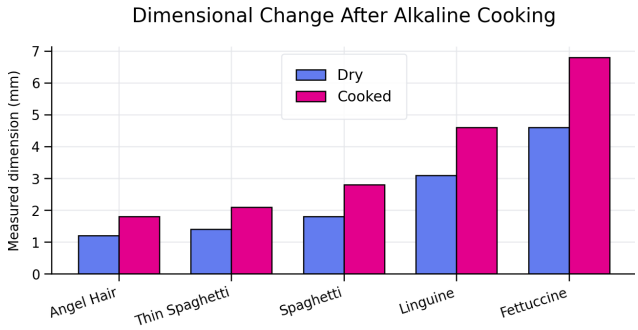


Figure 3. Dry and cooked measurements.

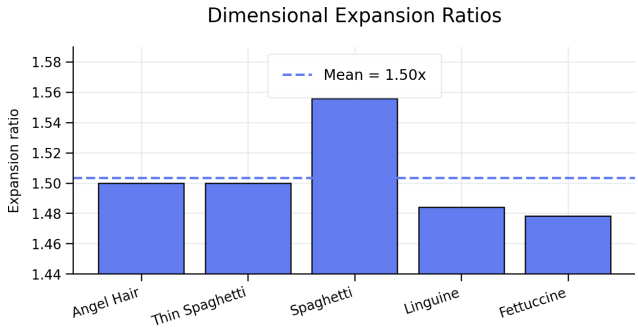


Figure 4. Expansion ratios; mean 1.50x.

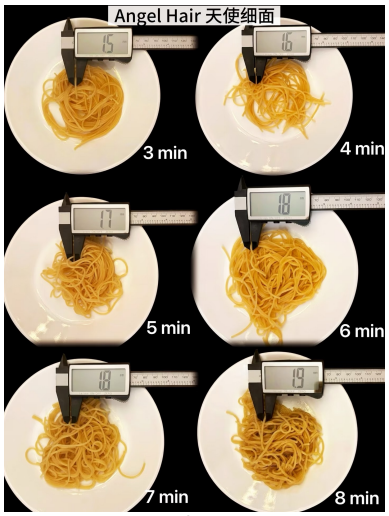


Figure 5. Angel Hair measurements from 3 to 8 minutes.

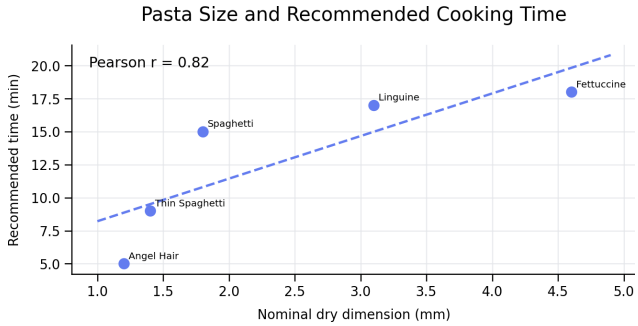


Figure 6. Descriptive relationship between nominal size and recommended time (Pearson r = 0.82).

4. 应用建议 / Culinary Applications

Angel Hair 和 Thin Spaghetti 更适合细面应用；标准 Spaghetti 适用范围较广；Linguine 和 Fettuccine 更适合炒面、拌面和宽面风格。该建议不表示这些产品与对应中式面条在原料或结构上完全相同。 / Finer formats suited thin-noodle applications; standard Spaghetti was the most versatile; Linguine and Fettuccine suited stir-fried, mixed, or broad-noodle styles. These are culinary suggestions, not claims of structural identity.



Figure 7. Original results and application suggestions.

5. 讨论 / Discussion

膨胀倍数集中于 1.48-1.56，但绝对增长和建议时间随规格增加。尺寸增长可以在感官最佳点之后继续，因此测量值与口感应共同考虑。 / Ratios clustered from 1.48 to 1.56, while absolute growth and recommended time increased with size. Swelling could continue beyond the preferred sensory endpoint, so measurement and texture should be considered together.

6. 局限性 / Limitations

未记录每种面条的测量根数、重复次数、卡尺校准、测量位置和夹持力度。混合使用宽度与直径限制了直接比较。五个异质产品的相关仅作描述。 / Strand count, replication, caliper calibration, measurement location, and jaw pressure were not documented. Mixed width and diameter measures limit direct comparison. The five-product correlation is descriptive only.

7. 结论 / Conclusion

在 4 g/L 小苏打条件下，建议时间为 5-18 分钟，平均尺寸膨胀约 1.50 倍。 / Under 4 g/L baking soda, recommended times ranged from 5 to 18 minutes and mean dimensional expansion was approximately 1.50x.

出版信息 / Publication Information

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